

JEE Mains 2019 Chapter wise Question Bank

Indefinite Integration – Questions

Q1

For $x^2 \neq n\pi + 1$, $n \in \mathbb{N}$ (the set of natural numbers), the integral

$$\int x \sqrt{\frac{2 \sin(x^2 - 1) - \sin 2(x^2 - 1)}{2 \sin(x^2 - 1) + \sin 2(x^2 - 1)}} dx \text{ is equal to:}$$

(1) $\log_e \left| \frac{1}{2} \sec^2(x^2 - 1) \right| + c$

(2) $\frac{1}{2} \log_e |\sec(x^2 - 1)| + c$

(3) $\frac{1}{2} \log_e \left| \sec^2 \left(\frac{x^2 - 1}{2} \right) \right| + c$

(4) $\log_e \left| \sec \left(\frac{x^2 - 1}{2} \right) \right| + c$

(where c is a constant of integration)

9 Jan Morning

Q2

If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx, (x \geq 0)$,

and $f(0) = 0$, then the value of $f(1)$ is:

(1) $-\frac{1}{2}$

(2) $-\frac{1}{4}$

(3) $\frac{1}{2}$

(4) $\frac{1}{4}$

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Q3

Let $n \geq 2$ be a natural number and $0 < \theta < \frac{\pi}{2}$

Then $\int \frac{(\sin^n \theta + \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta$ is equal to:

(1) $\frac{n}{n^2 - 1} \left(1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$

(2) $\frac{n}{n^2 + 1} \left(1 - \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$

(3) $\frac{n}{n^2 - 1} \left(1 + \frac{1}{\sin^{n-1} \theta} \right)^{\frac{n+1}{n}} + C$

(4) $\frac{n}{n^2 - 1} \left(1 - \frac{1}{\sin^{n+1} \theta} \right)^{\frac{n+1}{n}} + C$

(where C is a constant of integration)

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Q4

If $\int x^5 e^{-4x^3} dx = \frac{1}{48} e^{-4x^3} f(x) + C$, where C is a constant of integration, then $f(x)$ is equal to:

(1) $-2x^3 - 1$

(2) $-4x^3 - 1$

(3) $-2x^3 + 1$

(4) $4x^3 + 1$

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Q5

Indefinite Integration

If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x) \left(\sqrt{1-x^2} \right)^m + C$, for a suitable

chosen integer m and a function $A(x)$, where C is a constant of integration, then $(A(x))^m$ equals :

- (1) $\frac{-1}{27x^9}$ (2) $\frac{-1}{3x^3}$
(3) $\frac{1}{27x^6}$ (4) $\frac{1}{9x^4}$

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Q6

If $\int \frac{x+1}{\sqrt{2x-1}} dx = f(x) \sqrt{2x-1} + C$, where C is a constant

of integration, then $f(x)$ is equal to:

- (1) $\frac{1}{3}(x+1)$ (2) $\frac{2}{3}(x+2)$
(3) $\frac{2}{3}(x-4)$ (4) $\frac{1}{3}(x+4)$

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Q7

The integral $\int \cos(\log_e x) dx$ is equal to :

(where C is a constant of integration)

- (1) $\frac{x}{2} [\sin(\log_e x) - \cos(\log_e x)] + C$
(2) $x [\cos(\log_e x) + \sin(\log_e x)] + C$
(3) $\frac{x}{2} [\cos(\log_e x) + \sin(\log_e x)] + C$
(4) $x [\cos(\log_e x) - \sin(\log_e x)] + C$

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Q8

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The integral $\int \frac{3x^{13} + 2x^{11}}{(2x^4 + 3x^2 + 1)^4} dx$ is equal to:

(where C is a constant of integration)

- (1) $\frac{x^4}{6(2x^4 + 3x^2 + 1)^3} + C$ (2) $\frac{x^{12}}{6(2x^4 + 3x^2 + 1)^3} + C$
(3) $\frac{x^4}{(2x^4 + 3x^2 + 1)^3} + C$ (4) $\frac{x^{12}}{(2x^4 + 3x^2 + 1)^3} + C$

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Q9

$\int \frac{\sin -}{\sin -} dx$ is equal to :

(where c is a constant of integration.)

- (1) $2x + \sin x + 2 \sin 2x + c$
(2) $x + 2 \sin x + 2 \sin 2x + c$
(3) $x + 2 \sin x + \sin 2x + c$
(4) $2x + \sin x + \sin 2x + c$

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Q10

If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = xf(x)(1+x^6)^{\frac{1}{3}} + C$, where C is a

constant of integration, then the function $f(x)$ is equal to :

- (1) $\frac{3}{x^2}$ (2) $-\frac{1}{6x^3}$ (3) $-\frac{1}{2x^2}$ (4) $-\frac{1}{2x^3}$

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Q11

The integral $\int \sec^{2/3} x \operatorname{cosec}^{4/3} x dx$ is equal to:

- (1) $-3 \tan^{-1/3} x + C$ (2) $-\frac{3}{4} \tan^{-4/3} x + C$
(3) $-3 \cot^{-1/3} x + C$ (4) $3 \tan^{-1/3} x + C$
(Here C is a constant of integration)

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Indefinite Integration

Q12

If $\int e^{\sec x} (\sec x \tan x f(x) + (\sec x \tan x + \sec^2 x)) dx = e^{\sec x} f(x) + C$, then a possible choice of $f(x)$ is:

- (1) $\sec x + \tan x + \frac{1}{2}$ (2) $\sec x - \tan x - \frac{1}{2}$
(3) $\sec x + x \tan x - \frac{1}{2}$ (4) $x \sec x + \tan x + \frac{1}{2}$

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Q13

If $\int \frac{dx}{(x^2 - 2x + 10)^2} = A \left(\tan^{-1} \left(\frac{x-1}{3} \right) + \frac{f(x)}{x^2 - 2x + 10} \right) + C$ where C is a constant of integration, then :

- (1) $A = \frac{1}{54}$ and $f(x) = 3(x-1)$
(2) $A = \frac{1}{81}$ and $f(x) = 3(x-1)$
(3) $A = \frac{1}{27}$ and $f(x) = 9(x-1)$
(4) $A = \frac{1}{54}$ and $f(x) = 9(x-1)^2$

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Q14

If $\int x^5 e^{-x^2} dx = g(x) e^{-x^2} + c$, where c is a constant of integration, then $g(-1)$ is equal to :

- (1) -1 (2) 1
(3) $-\frac{5}{2}$ (4) $-\frac{1}{2}$

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Q15

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The integral $\int \frac{2x^3 - 1}{x^4 + x} dx$ is equal to :

(Here C is a constant of integration)

- (1) $\frac{1}{2} \log_e \left| \frac{x^3 + 1}{x^2} \right| + C$ (2) $\frac{1}{2} \log_e \frac{(x^3 + 1)^2}{|x^3|} + C$
(3) $\log_e \left| \frac{x^3 + 1}{x} \right| + C$ (4) $\log_e \frac{|x^3 + 1|}{x^2} + C$

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Q16

Let $\alpha \in (0, \pi/2)$ be fixed. If the integral

$$\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx = A(x) \cos 2\alpha + B(x) \sin 2\alpha + C, \text{ where } C \text{ is}$$

a constant of integration, then the functions $A(x)$ and $B(x)$ are respectively :

- (1) $x + \alpha$ and $\log_e |\sin(x + \alpha)|$
(2) $x - \alpha$ and $\log_e |\sin(x - \alpha)|$
(3) $x - \alpha$ and $\log_e |\cos(x - \alpha)|$
(4) $x + \alpha$ and $\log_e |\sin(x - \alpha)|$

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Indefinite Integration - Solutions

Q1

(3, 4) Consider the given integral

$$I = \int x \sqrt{\frac{2\sin(x^2-1) - 2\sin(x^2-1)\cos(x^2-1)}{2\sin(x^2-1) + 2\sin(x^2-1)\cos(x^2-1)}} dx$$

($\therefore \sin 2\theta = 2\sin\theta \cos\theta$)

$$\Rightarrow I = \int x \sqrt{\frac{1 - \cos(x^2-1)}{1 + \cos(x^2-1)}} dx$$

$$\Rightarrow I = \int x \left| \tan\left(\frac{x^2-1}{2}\right) \right| dx,$$

Now let $\frac{x^2-1}{2} = t \Rightarrow \frac{2x}{2} dx = dt$

$$\therefore I = \int |\tan(t)| dt = \ln|\sec t| + C$$

or $I = \ln \left| \sec\left(\frac{x^2-1}{2}\right) \right| + c = \frac{1}{2} \ln \left| \sec^2\left(\frac{x^2-1}{2}\right) \right| + c$

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Q2

(4) $f(x) =$

$$= \int \frac{5x^8 + 7x^6}{x^{14}(x^{-5} + x^{-7} + 2)^2} dx$$

$$= \int \frac{5x^{-6} + 7x^{-8}}{(2 + x^{-7} + x^{-5})^2} dx$$

Let $2 + x^{-7} + x^{-5} = t$

$$\Rightarrow (-7x^{-8} - 5x^{-6})dx = dt$$

$$\Rightarrow f(x) = \int \frac{-dt}{t^2} = \int -t^{-2} dt = t^{-1} + c$$

$$\Rightarrow f(x) = \frac{1}{2 + x^{-7} + x^{-5}} + c, f(0) = 0 \Rightarrow c = 0$$

$$\therefore f(1) = \frac{1}{4}$$

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Q3

(1) Let, $I = \int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta$

Let $\sin \theta = u$

$$\Rightarrow \cos \theta d\theta = du$$

$$\therefore I = \int \frac{(u^n - u)^{\frac{1}{n}}}{u^{n+1}} du$$

$$= \int \frac{\left(1 - \frac{1}{u^{n-1}}\right)^{\frac{1}{n}}}{u^n} du = \int u^{-n} (1 - u^{1-n})^{\frac{1}{n}} du$$

Let $1 - u^{1-n} = v$

$$\Rightarrow -(1-n)u^{-n} du = dv$$

$$\Rightarrow u^{-n} du = \frac{dv}{n-1}$$

$$\therefore I = \int v^{\frac{1}{n}} \cdot \frac{dv}{n-1} = \frac{1}{n-1} \cdot \frac{v^{\frac{1}{n}+1}}{\frac{1}{n}+1}$$

$$= \frac{n}{n^2-1} v^{\frac{n+1}{n}} + C = \frac{n}{n^2-1} \left(1 - \frac{1}{u^{n-1}}\right)^{\frac{n+1}{n}} + C$$

$$= \frac{n}{n^2-1} \left(1 - \frac{1}{\sin^{n-1} \theta}\right)^{\frac{n+1}{n}} + C$$

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Q4

Indefinite Integration

$$(2) \quad I = \int x^5 e^{-4x^3} dx$$

$$\text{Put } -4x^3 = \theta$$

$$\Rightarrow -12x^2 dx = d\theta$$

$$\Rightarrow x^2 dx = -\frac{d\theta}{12}$$

$$I = \int \frac{1}{48} \theta e^\theta d\theta = \frac{1}{48} [\theta e^\theta - e^\theta] + C$$

$$I = \frac{1}{48} e^{-4x^3} (-4x^3 - 1) + C$$

Then, by comparison

$$f(x) = -4x^3 - 1$$

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Q5

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$$(1) \quad A(x) \left(\sqrt{1-x^2} \right)^m + C = \int \frac{\sqrt{1-x^2}}{x^4} dx$$

$$= \int \frac{\sqrt{\frac{1}{x^2} - 1}}{x^3} dx$$

$$\text{Let } \frac{1}{x^2} - 1 = u^2$$

$$\Rightarrow -\frac{2}{x^3} = \frac{2u du}{dx}$$

$$\frac{dx}{x^3} = -u du$$

$$A(x) \left(\sqrt{1-x^2} \right)^m + C = \int (-u^2) du = -\frac{u^3}{3} + C$$

$$= -\frac{1}{3} \left(\frac{1}{x^2} - 1 \right)^{\frac{3}{2}} + C$$

$$= -\frac{1}{3} \cdot \frac{1}{x^3} \cdot (1-x^2)^{\frac{3}{2}} + C$$

$$= \frac{-1}{3x^3} \left(\sqrt{1-x^2} \right)^3 + C$$

Compare both sides,

$$\Rightarrow A(x) = -\frac{1}{3x^3} \text{ and } m = 3$$

$$\Rightarrow (A(x))^3 = \frac{-1}{27x^9}$$

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Q6

Indefinite Integration

(3) Let $I = \int \frac{x+1}{\sqrt{2x-1}} dx$

Put $\sqrt{2x-1} = t$

$\therefore 2x-1 = t^2 \Rightarrow dx = t dt$

$$I = \int \frac{(t^2+3)}{2} dt = \frac{t^3}{6} + \frac{3t}{2} + C$$

$$= \frac{(2x-1)^{\frac{3}{2}}}{6} + \frac{3}{2}(2x-1)^{\frac{1}{2}} + C$$

$$= \sqrt{2x-1} \left(\frac{x+4}{3} \right) + C$$

$$= f(x) \cdot \sqrt{2x-1} + C$$

Hence, $f(x) = \frac{x+4}{3}$

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Q7

(3) Let the integral, $I = \int \cos(\ln x) dx$

$$\Rightarrow I = \cos(\ln x) \cdot x - \int \frac{-\sin(\ln x)}{x} \cdot x dx$$

$$= x \cos(\ln x) + \int \sin(\ln x) dx$$

$$= x \cos(\ln x) + \sin(\ln x) \cdot x - \int \frac{\cos(\ln x)}{x} \cdot x dx$$

$$= x \cos(\ln x) + \sin(\ln x) \cdot x - I$$

$$\Rightarrow 2I = x(\cos(\ln x) + \sin(\ln x)) + C$$

$$\Rightarrow I = \frac{x}{2} [\cos(\ln x) + \sin(\ln x)] + C$$

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Q8

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(2) $I = \int \frac{3x^{13} + 2x^{11}}{(2x^4 + 3x^2 + 1)^4} dx = \int \frac{3x^{13} + 2x^{11}}{x^{16} \left(2 + \frac{3}{x^2} + \frac{1}{x^4} \right)^4} dx$

$$I = \int \frac{\frac{3}{x^3} + \frac{2}{x^5}}{\left(2 + \frac{3}{x^2} + \frac{1}{x^4} \right)^4} dx$$

Let $2 + \frac{3}{x^2} + \frac{1}{x^4} = t$, $-2 \left(\frac{3}{x^3} + \frac{2}{x^5} \right) dx = dt$

Then, $I = \int \frac{-\frac{dt}{2}}{t^4} = -\frac{1}{2} \frac{t^{-4+1}}{-4+1} + C$

$$I = \frac{-1}{2} \times \frac{1}{(-3)} \frac{1}{\left(2 + \frac{3}{x^2} + \frac{1}{x^4} \right)^3} + C$$

$$I = \frac{1}{6} \frac{x^{12}}{(2x^4 + 3x^2 + 1)^3} + C$$

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Q9

(3) $\int \frac{\sin\left(\frac{5x}{2}\right)}{\sin\left(\frac{x}{2}\right)} dx = \int \frac{2 \cos \frac{x}{2} \cdot \sin \frac{5x}{2}}{2 \cos \frac{x}{2} \cdot \sin \frac{x}{2}} dx$

$$= \int \frac{\sin 3x + \sin 2x}{\sin x} dx$$

$$= \int (3 - 4 \sin^2 x + 2 \cos x) dx$$

$[\because \sin 2x = 2 \sin x \cos x \text{ and } \sin 3x = 3 \sin x - 4 \sin^3 x]$

$$= \int (3 - 2(1 - \cos 2x) + 2 \cos x) dx$$

$$= \int (1 + 2 \cos x + 2 \cos 2x) dx$$

$$= x + 2 \sin x + \sin 2x + c$$

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Q10

Indefinite Integration

$$(4) \text{ Let, } \int \frac{dx}{x^3(1+x^6)^{\frac{2}{3}}} = \int \frac{dx}{x^7(1+x^{-6})^{\frac{2}{3}}}$$

$$\text{Put } 1+x^{-6}=t^3 \Rightarrow -6x^{-7}dx=3t^2dt \Rightarrow \frac{dx}{x^7}=\left(-\frac{1}{2}\right)t^2dt$$

$$\text{Now, } I = \int \left(-\frac{1}{2}\right) \frac{t^2dt}{t^2} = -\frac{1}{2}t + C$$

$$= -\frac{1}{2}(1+x^{-6})^{\frac{1}{3}} + C = -\frac{1}{2} \frac{(1+x^6)^{\frac{1}{3}}}{x^2} + C$$

$$= -\frac{1}{2x^3} x(1+x^6)^{\frac{1}{3}} + C$$

$$\text{Hence, } f'(x) = -\frac{1}{2x^3}$$

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Q11

$$(1) I = \int \sec^{\frac{2}{3}} x \cdot \operatorname{cosec}^{\frac{4}{3}} x dx$$

$$I = \int \frac{\sec^2 x dx}{\tan^{\frac{4}{3}} x}$$

$$\text{Put } \tan x = z$$

$$\Rightarrow \sec^2 x dx = dz$$

$$\Rightarrow I = \int z^{-\frac{4}{3}} \cdot dz = \frac{z^{-\frac{1}{3}}}{(-\frac{1}{3})} + C \Rightarrow I = -3(\tan x)^{-\frac{1}{3}} + C$$

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Q12

(1) Given,

$$\int e^{\sec x} \left(\sec x \tan x f'(x) + (\sec x + \tan x + \sec^2 x) \right) dx = e^{\sec x} f(x) + C \quad \dots(1)$$

$$\therefore \int e^{g(x)} \left((g'(x)f'(x)) + f'(x) \right) dx = e^{g(x)} \times f(x) + C$$

Our comparing above equation by equation (1),

$$f(x) = \int (\sec x \tan x + \sec^2 x) dx$$

$$\therefore f(x) = \sec x + \tan x + C$$

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Q13

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$$(1) \text{ Let } I = \int \frac{dx}{(x^2 - 2x + 10)^2} = \int \frac{dx}{((x-1)^2 + 9)^2}$$

$$\text{Let } (x-1)^2 = 9 \tan^2 \theta \quad \dots(1)$$

$$\Rightarrow \tan \theta = \frac{x-1}{3}$$

After differentiating equation ... (1), we get

$$2(x-1)dx = 18 \tan \theta \sec^2 \theta d\theta$$

$$\therefore I = \int \frac{18 \tan \theta \sec^2 \theta d\theta}{2 \times 3 \tan \theta \times 81 \sec^4 \theta}$$

$$I = \frac{1}{27} \int \cos^2 \theta d\theta = \frac{1}{27} \times \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$I = \frac{1}{54} \left\{ \theta + \frac{\sin 2\theta}{2} \right\} + c$$

$$I = \frac{1}{54} \left[\tan^{-1} \left(\frac{x-1}{3} \right) + \frac{1}{2} \times \frac{2 \left(\frac{x-1}{3} \right)}{1 + \left(\frac{x-1}{3} \right)^2} \right] + c$$

$$I = \frac{1}{54} \left[\tan^{-1} \left(\frac{x-1}{3} \right) + \frac{3(x-1)}{x^2 - 2x + 10} \right] + c$$

$$\text{Compare it with } A \left[\tan^{-1} \left(\frac{x-1}{b} \right) + \frac{f(x)}{x^2 - 2x + 10} \right] + c,$$

$$\text{we get: } A = \frac{1}{54} \text{ and } f(x) = 3(x-1)$$

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Q14

$$(3) \text{ Let, } 1 = \int x^2 \cdot e^{-x^2} dx$$

$$\text{Put } -x^2 = t \Rightarrow -2x dx = dt$$

$$1 = \int \frac{t^2 \cdot e^t dt}{(-2)} = \frac{-1}{2} e^t (t^2 - 2t + 2) + c$$

$$\therefore g(x) = \frac{-1}{2} (x^4 + 2x^2 + 2) \Rightarrow g(-1) = \frac{-5}{2}$$

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Q15

(3) Given integral, $I = \int \frac{(2x^3 - 1)dx}{x^4 + x} = \int \frac{(2x - x^{-2})dx}{x^2 + x^{-1}}$

Put $x^2 + x^{-1} = u \Rightarrow (2x - x^{-2})dx = du$

$$\Rightarrow I = \int \frac{du}{u} = \log |u| + c = \log |x^2 + x^{-1}| + c$$

$$= \log \left| \frac{x^3 + 1}{x} \right| + c$$

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Q16

(2) Given integral

$$\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx = \int \frac{\sin(x + \alpha)}{\sin(x - \alpha)} dx$$

Let $x - \alpha = t \Rightarrow dx = dt$

$$= \int \frac{\sin(t + 2\alpha)}{\sin t} dt = \int [\cos 2\alpha + \sin 2\alpha \cdot \cot t] dt$$

$$= \cos 2\alpha \cdot t + \sin 2\alpha \cdot \log |\sin t| + c$$

$$= (x - \alpha) \cdot \cos 2\alpha + \sin 2\alpha \cdot \log |\sin(x - \alpha)| + c$$

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